MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE, BA-5800/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the lithium sulfur dioxide non-rechargeable battery described herein shall consist of this document and the latest issue of MIL-B-49430(ER).

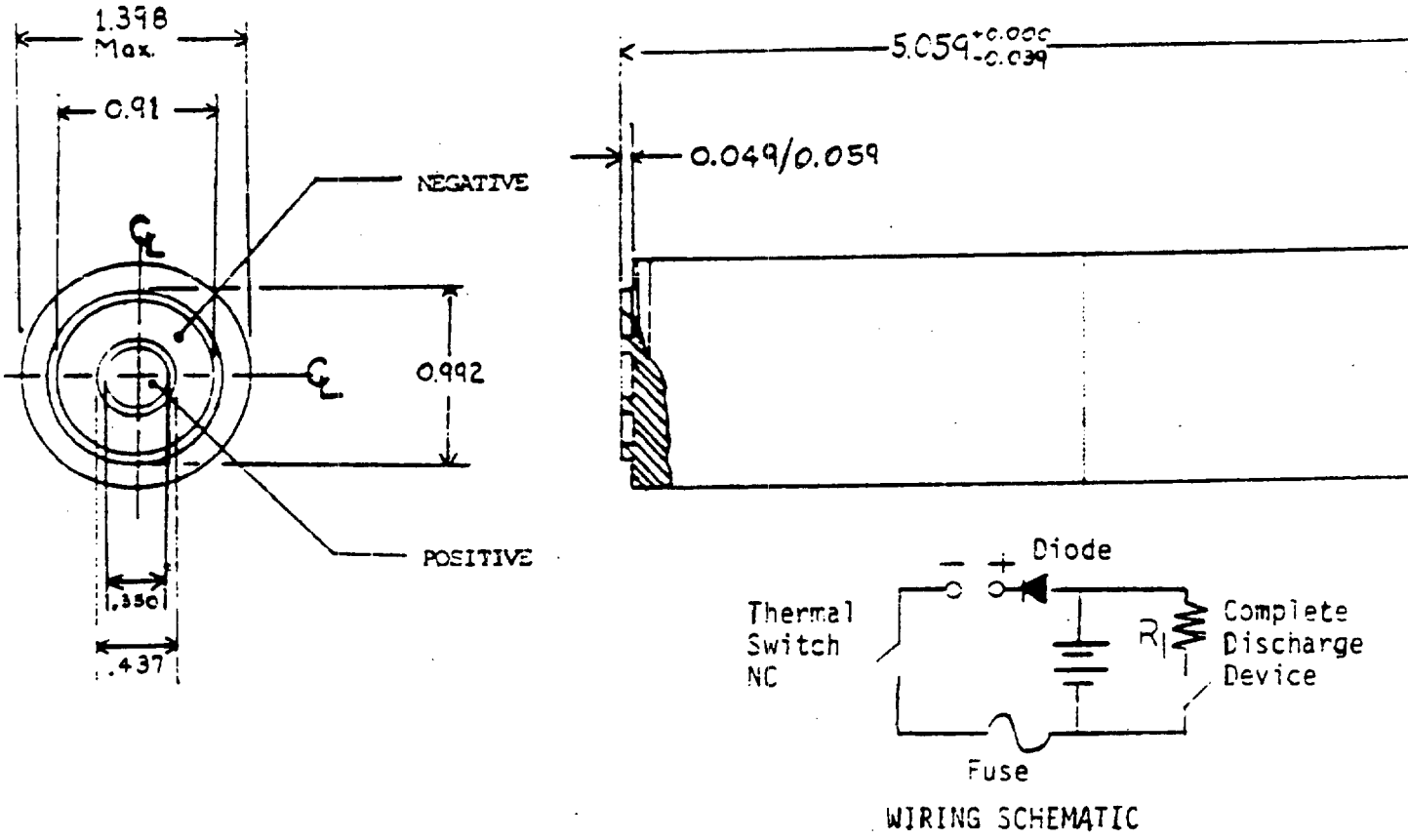


Figure 1. Battery dimensions and wiring schematics.

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NOTES:

1. Cell dimensions are in inches.

2. Tolerances are as follows:

X.XX	±	0.014
X.XXX	±	0.004

3. Electrical contacts are 0.01 inches thick.

4. R_1 value shall be between 60 ohms and 84 ohms and have a power rating of 0.5 watt.

5. Socket wiring schematic to be printed or embossed on jacket in suitable location near the electrical contacts.

EQUIREMENTS

Dimensions and configuration: See Figure 1.

Maximum voltage: 6.10 volts.

Minimum voltage for capacaity: 4.00 volts.

The battery shall contain a 2.25 ampere non-replaceable slow blow fuse wired in series with the cells in the negative leg.

The battery shall contain a diode wired in series. See Paragraph 3.5.5.2 of Basic Specification.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery as described in paragraphs 3.5.5 and 3.5.5.1 of the Basic Specification.

Weight (maximum): 0.48 pound (220 grams).

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para. 3.5.5.3 of the basic specification.

Capacity tests: When the battery is tested in accordance with the method of examination and tests of this specification, the minimum capacity requirements shall be as specified below:

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Capacity test	Battery Capacity Requirements in hours to 4.00 volts
I	3.0
L	1.6
H	5.6
HT	5.4
LT	1.4
T	2.8
D	2.8
ILR	1000

Initial voltage delay: During the I,H,HT,T,D, and ILR discharge test, initial voltage below 4.00 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests, initial voltages below 3.80 volts cannot exceed a 60.0 second duration.

METHODS OF EXAMINATION AND TESTS:

Capacity test:

(1.) Storage: Details on storage conditions for all capacity tests are specified in the basic specification (see 4.8.9).

(2.) Discharge:

Cells: Discharge at 2.0 amperes constant current at $70\pm5^{\circ}$ for a minimum of 2.7 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 amperes constant current for 5.4 hours into voltage reversal.

Batteries: For all capacity tests excepts the H,HT, and ILR, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. There shall be no venting, fire or explosion when the batteries are discharged. For the ILR test a fixed resistance of 1050 ohms shall be used to cut-off voltage of 4.00 volts or 1000 hours.

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Closed circuits voltage test: Load resistance of 5 ohms shall be used. Voltage shall be above 4.00 volts within 5 seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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